Algorithm 1 Fitting the dial (offline, BDD100K train only)

Require: labelled train frames $\mathcal{D}_{\text{train}}$; frozen detector f

for all $x \in \mathcal{D}_{\text{train}}$ **do**

  $(D(x), h) \leftarrow f(x)$ ▷ floor 0.001, NMS 0.5, ≤ 300 boxes; keep per-box outputs

  $y(d) \leftarrow$ greedy IoU ≥ 0.5 match of $d \in D(x)$ to labels ▷ grouped classes

Option A: sort scores; keep ends of equal-score blocks; cumulative real and false counts

  $\tau(\cdot) \leftarrow$ corners of the lower convex hull ▷ all λ at once

Option B: $\varphi(d) \leftarrow [\text{context}(d, D(x)),\ \text{per-box}(h(d))]$ ▷ PCA fitted on train

  $g \leftarrow$ gradient boosting on $\{(\varphi(d), y(d))\}$ ▷ log-loss

  $p_{\text{oof}} \leftarrow$ 5-fold out-of-fold probabilities, folds by near-duplicate group

for all β **do**

  $\lambda_{\text{eq}}(\beta) \leftarrow (1 + \beta^2)/F^*_\beta - 1$ ▷ F-score map, F^*_β on train

check calibration near every cut $1/(1 + \lambda)$ and fold stability; adopt only if the pre-registered criteria hold

return $\tau(\cdot)$, g (with PCA), $\lambda_{\text{eq}}(\cdot)$

Algorithm 2 Applying the dial (online, every camera frame)

Require: frame x; setting λ (or β); option $\in \{\text{A}, \text{B}\}$

if β is given **then**
  $\lambda \leftarrow \lambda_{\text{eq}}(\beta)$ ▷ F-score map
if a budget α is certified for condition c **then**
  $\lambda \leftarrow \max(\lambda, \hat{\lambda}_c)$ ▷ planned: CRC over λ
$(D, h) \leftarrow f(x)$ ▷ one detector pass
for all $d \in D$ **do**
  if option A **then**
    keep d iff $s(d) \geq \tau(\lambda)$
  else
    $p(d) \leftarrow \sigma(g(\varphi(d)))$; keep d iff $p(d) \geq 1/(1+\lambda)$ ▷ cost-optimal cut
$\widehat{\text{FA}} \leftarrow \sum_{d \text{ kept}} (1 - p(d))$ ▷ expected false alarms, no labels needed
return kept detections, $\widehat{\text{FA}}$